Machine Learning-Based Classification of Household Poverty in Rwanda Using EICV7 Data

NISR 2026 Big Data Hackathon — Participation Formative

Track	Financial Inclusion & Poverty Reduction
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1. Introduction

Poverty remains an important development challenge because households differ in their access to adequate living conditions, basic services, assets, and economic resources. This project uses the Rwanda Integrated Household Living Conditions Survey (EICV7), a nationally representative cross-sectional survey conducted from October 2023 to October 2024 by the National Institute of Statistics of Rwanda (NISR). The EICV7 documentation reports 15,054 successfully interviewed households out of 15,066 targeted households.

Problem statement. The project investigates whether socioeconomic and household characteristics can classify households as poor or non-poor.
Research question: Can socioeconomic and household characteristics be used to classify households as poor or non-poor using EICV7 data? Since EICV7 is cross-sectional, the project classifies observed poverty status rather than forecasting future poverty.

Dataset. The household dataset *CS_S01_S5_S7_Household* contains 15,054 cases and 112 variables. It includes household characteristics such as dwelling conditions, rooms, ownership, water, electricity, internet, sanitation, construction materials, household shocks and livestock. The documentation also identifies *pov_jan* (Poverty Rate), *poverty* (Welfare Categories), *epov_jan* (Extreme Poverty), and *quintile* (Quintiles of real consumption per ae) as poverty-related variables.

2. Methodology

Target and leakage control. The primary target is *pov_jan*, converted to a binary outcome: 0 = non-poor and 1 = poor. Direct poverty indicators such as *poverty*, *quintile*, and *epov_jan*, together with identifiers and other direct outcome-derived fields, are excluded from predictors to reduce target leakage.

Preprocessing. The workflow loads the EICV7 Stata household file, inspects variable types and missingness, removes identifiers/leakage-prone variables, and separates numerical and categorical predictors. Numerical missing values are median-imputed; categorical missing values use the most-frequent category. Categorical variables are one-hot encoded and numerical variables are standardized where required. A stratified train/test split preserves the class distribution.

Model architecture and justification

Logistic Regression is an interpretable linear baseline that estimates the probability of the poor class. **Random Forest** combines decision trees and captures nonlinear relationships and interactions while providing feature importance. **XGBoost** sequentially builds gradient-boosted decision trees to reduce previous errors and model complex nonlinear patterns in tabular socioeconomic data.

Evaluation. Models are compared using accuracy, precision, recall, F1-score and ROC-AUC. Recall is important because failing to identify a poor household is a meaningful error. F1 balances precision and recall, while ROC-AUC measures ranking performance across thresholds. A confusion matrix is used to inspect false positives and false negatives.

Implementation status. The project is implemented in Python and prepared for Google Colab. The notebook extracts the uploaded EICV7 microdata ZIP, locates *CS_S01_S5_S7_Household.dta*, loads it with pandas, performs preprocessing, trains the three baseline classifiers, evaluates them, and saves model artifacts. For the formative stage, the code is structured to run end-to-end with a short placeholder training run; full optimization and final training can be performed after the formative submission.

TensorBoard. TensorBoard logging is included so experiment information and metrics can be visualized. The logging setup can be extended during final development to record additional hyperparameters, metrics, and model comparisons.

Ethical and practical consideration. This is an analytical classification experiment, not an automated system for denying benefits or making high-stakes decisions about individual households. Results should be interpreted in light of the survey design, data limitations and possible class imbalance.

3. Expected Contribution

The project will provide a reproducible machine-learning pipeline for poverty classification using Rwanda's EICV7 household data. It will compare an interpretable linear baseline with tree-based models, examine household characteristics associated with classification, and establish a baseline that can later be improved through hyperparameter tuning, class-imbalance handling, feature selection and additional validation. The work directly supports the Financial Inclusion & Poverty Reduction track.

4. Implementation Checklist

Component	Status

EICV7 household microdata loaded	Completed
Binary poverty target defined	Completed
Leakage/identifier removal	Completed
Preprocessing pipeline	Completed
Logistic Regression / Random Forest / XGBoost	Implemented
Evaluation + confusion matrix	Implemented
TensorBoard setup	Implemented
Full optimization / final training	To be completed

Appendix: Submission Links

Code / Google Colab or GitHub	https://colab.research.google.com/drive/1fzF94J-uvUt4PSoiB26WZDVLh86Z0T6-?usp=sharing
Loom registration recording	https://www.loom.com/share/6482c0f3938c4f098365dbd40575b776